

Nathan Jacobson

njacobson604@gmail.com | github.com/grqphical | nathanjacobson.ca

WORK EXPERIENCE

Tae Kwon Do Instructor

September 2025 — December 2025

Transcending Martial Arts

Port Moody, BC

- Instructed children aged 4-14 martial arts techniques such as strikes, blocks, patterns, and weapons training
- Led a group of volunteer assistant instructors to aid with class management
- Developed structured lesson plans based on the class' age and skill levels

Information Officer

April 2025

Elections Canada

Port Moody, BC

- Confirmed voter identification when they entered the polling station
- Directed voters to the correct stations and assisted voters with physical disabilities
- Recorded vote counts after polls had closed

VOLUNTEER EXPERIENCE

Assistant Tae Kwon Do Instructor

June 2022 — August 2023

Transcending Martial Arts

Port Moody, BC

- Aided instructors in teaching children aged 4-9 martial arts techniques
- Managed the class, ensuring students were focused
- Helped students who were struggling with following the lesson

Summer Camp Volunteer

July 2023

Simon Fraser University Summer Camps

Burnaby, BC

- Assisted instructors in running chemistry and engineering camps for children ages 5-6
- Supervised groups of children during structured activities, lunch breaks, and free time

PROJECTS

interchange (<https://github.com/grqphical/interchange>)

August 2025 — December 2025

- Created a reverse proxy in Go that supports proxying requests, static file hosting, HTTPS, and IP blacklisting/whitelisting

Catan AI (<https://github.com/grqphical/catan-ai>)

January 2026 — Present

- Currently Building a web app with React, Python, and FastAPI that plays the board game Settlers of Catan
- Uses algorithms to determine the best initial town placements, the best move to make on it's turn, and many other parts of the game
- Players input their board setup in the web app initially, then every turn input the dice roll and what other players did that turn

Homelab

October 2025 — Present

- Created a Debian-based Linux server using a spare laptop, ensuring 24/7 uptime
- Deployed a suite of services using Docker, including a remote file browser, a DNS server, a Minecraft server, and a server dashboard
- Configured a mesh WireGuard VPN to securely access the server

SKILLS

- **Programming Languages:** Python, Go, C, C++, HTML/CSS/JS, SQL
- **Libraries/Tools:** Linux, numpy, pandas, FastAPI, React, SQLite, matplotlib
- **Non-Programming Skills:** Image Editing, MS Office, Communication, Ability to work autonomously and in groups

EDUCATION

Simon Fraser University

Burnaby, BC

Bachelor's of Science, Computing Science

September 2025 — Present

- Relevant course work: Data Structures and Algorithms, Systems Programming, Linear Algebra, and Discrete Mathematics

Heritage Woods Secondary

Port Moody, BC

BC Dogwood Dipolma (High School Diploma)

September 2021 — June 2025

- Awarded Top Producing Grade 12 Student and BC Achievement Scholarship
- Graduated with Honour Roll with Distinction

EXTRA-CURRICULARS

Tae Kwon Do

September 2017 — Present

- Achieved a 1st Degree Black Belt